Supporting Information for: Early leafout extends calendar but not thermal growing seasons

July 16, 2026

1 Tables

Species	functional group	n
**Acer pensylvanicum*	tree	*NA*
**Acer spicatum*	tree	*NA*
Alnus incana	shrub	31
Amelanchier canadensis	shrub	6
Aronia melanocarpa	shrub	12
Betula alleghaniensis	tree	24
Betula papyrifera	tree	13
Betula populifolia	tree	24
Diervilla lonicera	shrub	16
Myrica gale	shrub	15
**Quercus alba*	tree	*NA*
**Quercus rubra*	tree	*NA*
Sambucus racemosa	shrub	11
Sorbus americana	shrub	5
Spiraea alba	shrub	19
Spiraea tomentosa	shrub	21
**Vaccinium myrtilloides*	shrub	*NA*
Viburnum cassinoides	shrub	25

Table S1: Common garden focal species. In 2017, plantings were randomized between 16 plot blocks. Individuals that were too small to survive outside were maintained in the growth facilities for an additional year and out-planted in the early spring of 2018. Plots were divided between tree plots which included species *Acer pensylvanicum*, *Amelanchier canadensis*, *Alnus incana*, *Betula papyrifera*, *Betula populifolia*, *Beluta alleghaniensis*, *Quercus alba*, and *Quercus rubra* and shrub plots which included the remaining species and shade cloth. * indicates species that were planted but not included in analyses due to low survivorship.

2 Figures

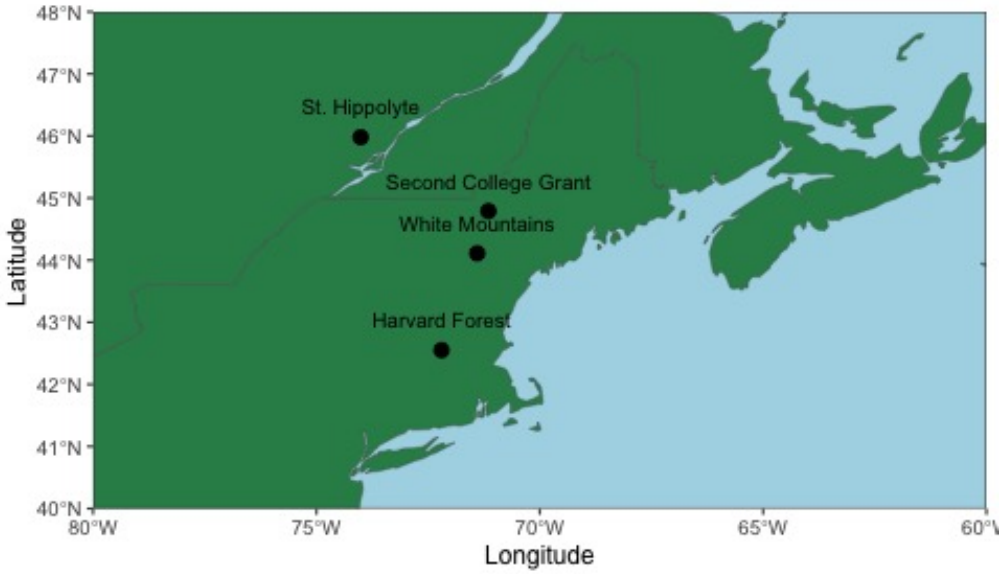


Figure S1: A map of the four site sampled in this study

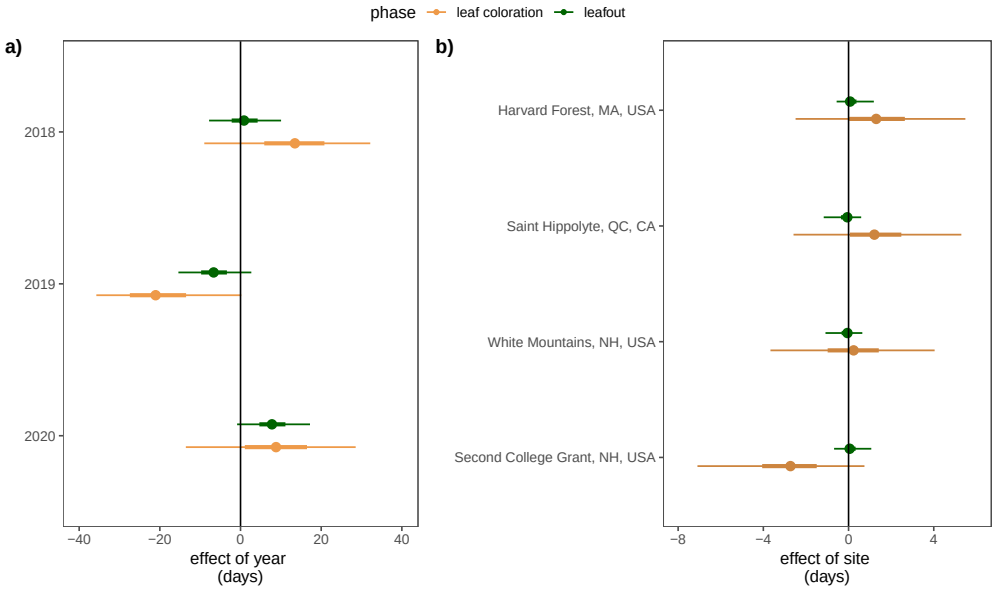


Figure S2: Difference in leafout and leaf coloration partitioned between years (a) and populations (b). Points represent the median effect size estimate, and thick and thin bars the 50% and 90% uncertainty intervals.

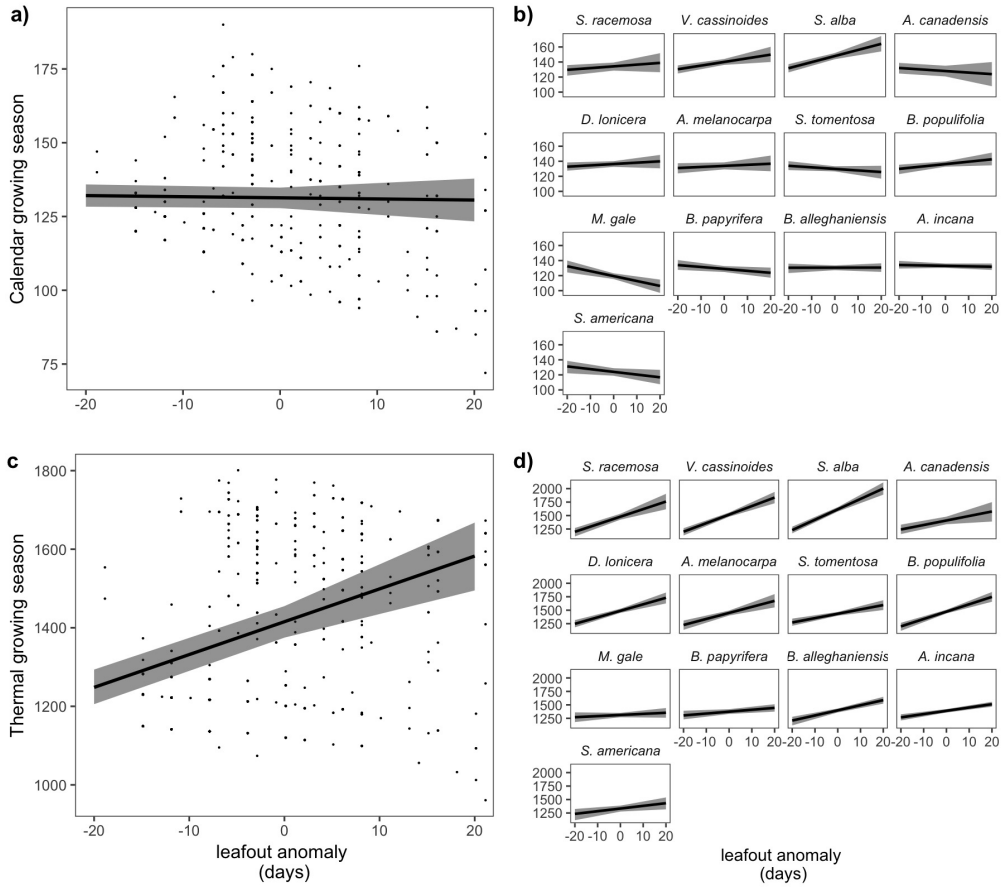


Figure S3: The relationship between the day of leafout and growing season length (defined by leaf coloration) differs between the calendar growing season and the thermal growing season. Later leafout did not affect calendar growing season (a) but this pattern varied across species in our study (b). Increasingly later leafout resulted in a longer thermal growing season (c) though this effect was stronger for species that typically leafout earlier in the season—panels in c) display in the typical order of leafout among species. The trend lines represent the median effect of leafout timing on growing season length and shaded bars the 90% uncertainty interval. Points in a), and c) represent the raw data.

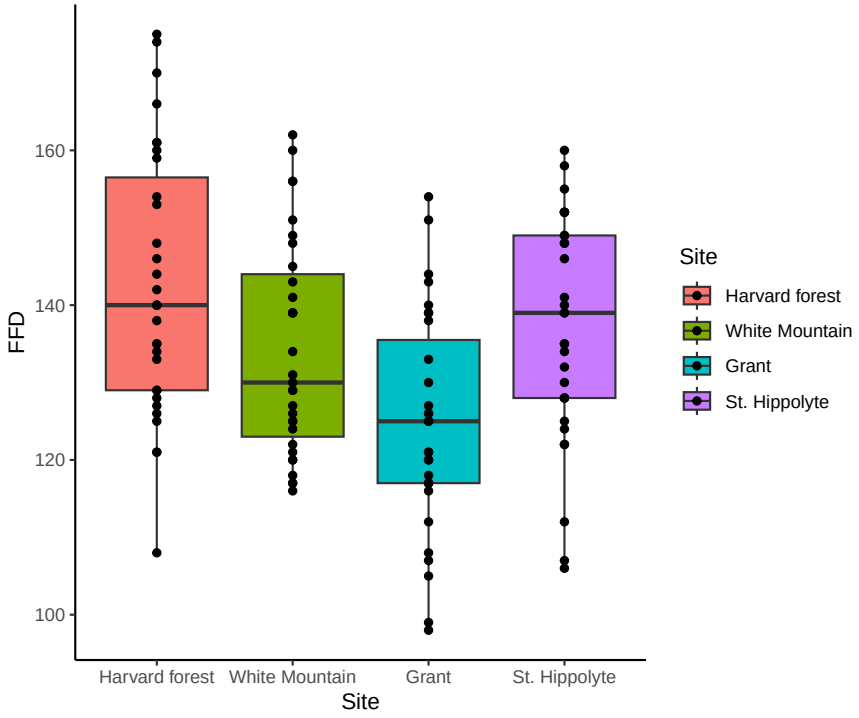


Figure S4: Differences in frost free days (FFD) between sites. Each point represents the number of days between the 90th percentile of last spring frost days and the 90th percentile of first fall frost days for a given year between 1980 and 2010. To calculate FFD, we used the long-term gridded esimates of minimum temperature from Daymet (version 4, **?**).